Linear Statistical Models

B. Statistical Data Science 2nd Year   Indian Statistical Institute

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Linear Algebra Revision 2

Exercise 1. In this exercise, we will make precise some statements made in the class about bases. Recall that a collection of vectors $\{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ is a basis for the vector space $\mathcal{V}$ if

(i) $\mathcal{V} = \mathrm{span}\{\mathbf{b}_1, \dots, \mathbf{b}_p\}$, i.e., any $\mathbf{x} \in \mathcal{V}$ can be written as $\mathbf{x} = x_1\mathbf{b}_1 + \cdots + x_p\mathbf{b}_p$.

(ii) $\mathbf{b}_1, \dots, \mathbf{b}_p$ are linearly independent, i.e., $\alpha_1\mathbf{b}_1 + \cdots + \alpha_p\mathbf{b}_p = \mathbf{0}$ implies $\alpha_1 = \cdots = \alpha_p = 0$.

(a) Show that if $\{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ is a basis for $\mathcal{V}$, then any collection of vectors $\{\mathbf{b}_1, \dots, \mathbf{b}_p, \mathbf{x}\}$ for $\mathbf{x} \in \mathcal{V}$ will be linearly dependent. Deduce that any proper superset of $\{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ is necessarily linearly dependent. This shows that a basis is **a** maximal collection of linearly independent vectors in $\mathcal{V}$.

Note: A basis is "a" (and not "the") maximal collection of linearly independent vectors!!

(b) Show that if $\{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ is a basis of $\mathcal{V}$, then $\{\mathbf{b}_1, \dots, \mathbf{b}_{p-1}\}$ DOES NOT span $\mathcal{V}$. Hence, deduce that any proper subset of a basis does not span the vector space under consideration. This shows that a basis is **a** minimal collection of vectors that span the vector space $\mathcal{V}$.

(c) Suppose that $\{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ is a basis for $\mathcal{V}$. Let $\mathbf{x} = x_1\mathbf{b}_1 + \cdots + x_i\mathbf{b}_i + \cdots + x_p\mathbf{b}_p$ be a vector in $\mathcal{V}$ such that $x_i \neq 0$. Show that $\{\mathbf{b}_1, \dots, \mathbf{b}_{i-1}, \mathbf{x}, \mathbf{b}_{i+1}, \dots, \mathbf{b}_p\}$ is also a basis for $\mathcal{V}$. So, there can (and will) exist multiple bases for the same vector space. Basis of a vector space is NOT unique.

(d) Show that if we are given a collection of linearly independent vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_d\} \subset \mathcal{V}$, then we can always find a basis for $\mathcal{V}$ that contains $\mathbf{v}_1, \dots, \mathbf{v}_d$. In other words, we can always **extend** a collection of linearly independent vectors to form a basis.

Hint: *Use the replacement trick from the previous part.*

(e) Suppose that $\{\mathbf{b}_1, \dots, \mathbf{b}_p\}$ and $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ are both bases for $\mathcal{V}$. Then, show that $p = k$. This shows that any basis of a vector space contains the same number of elements. Hence, the dimension of a vector space is unambiguously defined.

(f) Show that for a p-dimensional vector space $\mathcal{V}$, any collection $\{\mathbf{v}_1, \dots, \mathbf{v}_p\} \subset \mathcal{V}$ of p linearly independent vectors is a basis for $\mathcal{V}$.

Next, we will formally define inner-products and norms. The following exercises are mathematically interesting, but will not be very helpful in the course.

Exercise 2. Let $\mathcal{V}$ be a (real) vector space (i.e., a vector space defined over $\mathbb{R}$ as in Series 1). A map $\langle \cdot, \cdot \rangle : \mathcal{V} \times \mathcal{V} \to \mathbb{R}$ is called an inner-product if

(i) $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle \ \forall\, \mathbf{x}, \mathbf{y} \in \mathcal{V}$.

(ii) $\langle a \cdot \mathbf{x} + b \cdot \mathbf{y}, \mathbf{z} \rangle = a\langle \mathbf{x}, \mathbf{z} \rangle + b\langle \mathbf{y}, \mathbf{z} \rangle \ \forall\, \mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathcal{V}$ and $a, b \in \mathbb{R}$.

(iii) $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0 \ \forall\, \mathbf{x} \in \mathcal{V}$, and "=" holds if and only if $\mathbf{x} = \mathbf{0}$.

We also say that $\langle \cdot, \cdot \rangle$ is an inner-product on $\mathcal{V}$. (i) implies that $\langle \cdot, \cdot \rangle$ is *symmetric* in its arguments. (ii) implies that $\langle \cdot, \cdot \rangle$ is linear in each of the arguments. Such a function is also called *bilinear*. (iii) is known as *positive definiteness*.

Verify that the "inner-product" defined in the class:

$$\mathbf{x}^\top \mathbf{y} = \sum_{i=1}^{n} x_i y_i \text{ for } \mathbf{x} = (x_1, \ldots, x_n)^\top \text{ and } \mathbf{y} = (y_1, \ldots, y_n)^\top$$

is indeed an inner-product on $\mathbb{R}^n$ in the above sense.

NOT ALL vector spaces have an inner-product. But SOME do. In fact, the same vector space can have more than one inner-product. Show that the following are inner-products.

(a) For any $\rho \in (0,1)$, $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^{n} x_i y_i + \rho \sum\sum_{1 \le i \ne j \le n} x_i y_j$ for $\mathbf{x} = (x_1, \ldots, x_n)^\top, \mathbf{y} = (y_1, \ldots, y_n)^\top \in \mathbb{R}^n$.

(b) For any symmetric, positive definite matrix A, $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathrm{A} \mathbf{y}$ for $\mathbf{x} = (x_1, \ldots, x_n)^\top, \mathbf{y} = (y_1, \ldots, y_n)^\top \in \mathbb{R}^n$.

(c) $\langle f, g \rangle = \int f(x) g(x) \, \mathrm{d}x$ for $f, g \in \mathcal{V} = \left\{ f : \mathbb{R} \to \mathbb{R} \text{ s.t. } \int f^2(x) \, \mathrm{d}x < \infty \right\}$.

Note: Check also that $\mathcal{V}$ is a vector space. Can you use the subspace argument?

(d) $\langle X, Y \rangle = \mathbb{E}(XY)$ for $X, Y \in \mathcal{L}_2 = \{X : \mathrm{Var}(X) < \infty\}$.

Note: Do you see any issue here with property (iii)? What is the $\mathbf{0}$ vector in $\mathcal{L}_2$?

Note: Is $\langle X, Y \rangle = \mathrm{Cov}\{X, Y\}$ an inner-product on $\mathcal{L}_2$? What's the issue?

Vector spaces that have an inner-product defined on them are called *inner-product spaces*.

Exercise 3. Let $\mathcal{V}$ be a real vector space. A map $\|\cdot\| : \mathcal{V} \to \mathbb{R}_+$ is called a norm if

(i) $\|a \cdot \mathbf{x}\| = |a| \|\mathbf{x}\| \ \forall \mathbf{x} \in \mathcal{V}$ and $a \in \mathbb{R}$ (homogeneity).

(ii) $\|\mathbf{x} + \mathbf{y}\| \le \|\mathbf{x}\| + \|\mathbf{y}\| \ \forall \mathbf{x}, \mathbf{y} \in \mathcal{V}$ (triangle inequality).

(iii) $\|\mathbf{x}\| \ge 0 \ \forall \mathbf{x} \in \mathcal{V}$, with "=" holding if and only if $\mathbf{x} = 0$ (positive definiteness/positivity).

Show that the usual/Euclidean norm defined in the class

$$\|\mathbf{x}\| = \sqrt{\sum_{i=1}^{n} x_i^2} \text{ for } \mathbf{x} = (x_1, \ldots, x_n)^\top \in \mathbb{R}^n$$

is indeed a norm according to the above definition.

As with inner-products, not all vector spaces have a norm, but some do. Spaces that have a norm defined on them are called normed vector spaces or normed linear spaces. Show that the following are norms.

(a) $\|\mathbf{x}\|_1 = \sum_{i=1}^{n} |x_i|$ for $\mathbf{x} = (x_1, \ldots, x_n)^\top \in \mathbb{R}^n$. This is called the ℓ_1-norm.

(b) For $p \ge 1$, $\|\mathbf{x}\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}$ for $\mathbf{x} = (x_1, \ldots, x_n)^\top \in \mathbb{R}^n$. This is called the ℓ_p-norm.

(c) $\|\mathbf{x}\|_\infty = \max\{|x_1|, \dots, |x_n|\}$ for $\mathbf{x} = (x_1, \dots, x_n)^\top \in \mathbb{R}^n$. This is called the ℓ_∞-norm.

Show that for any $\mathbf{x} = (x_1, \dots, x_n)^\top \in \mathbb{R}^n$, $\|\mathbf{x}\|_p \to \|\mathbf{x}\|_\infty$ as $p \to \infty$. Hence the notation ℓ_∞.

(d) $\|X\|_1 = \mathbb{E}|X|$ for $X \in \mathcal{L}_1 = \{X : \mathbb{E}|X| < \infty\}$. This is called the $\mathcal{L}_1$ norm.

(e) $\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$ for $\mathbf{x} \in \mathcal{V}$, where $\langle \cdot, \cdot \rangle$ is an inner-product on $\mathcal{V}$. Thus, any inner-product induces a norm. The converse is NOT always true.

Note: Inner-products and norms play an important role in linear algebra. A norm can be viewed as the "size" of a vector. Inner-product, on the other hand, measures the strength of "dependence" between two vectors. Norms and inner-products are connected via the Cauchy-Schwarz inequality:

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \le \|\mathbf{x}\| \, \|\mathbf{y}\| \qquad \forall \, \mathbf{x}, \mathbf{y} \in \mathcal{V},$$

where "=" holds if and only if $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent. Here, $\|\cdot\|$ is the norm induced by the inner product $\langle \cdot, \cdot \rangle$, i.e., $\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$. This is used to define the "angle" between two vectors as

$$\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\| \, \|\mathbf{y}\| \cos(\theta) \quad \Leftrightarrow \quad \theta = \cos^{-1}\left(\frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \, \|\mathbf{y}\|} \right).$$

Observe that the angle is well-defined due to the Cauchy-Schwarz inequality. Do you see anything familiar from Exercise 3, part (d)?